~~0.12 nanometers or 0.12×10^{-9}~~
~~1.2×10^{-10}~~
~~1.2E-10 m~~

~~100~~
~~12×10^{1}~~
~~1×10^{2}~~

4.

$\frac{1\,L}{0.803\,kg}$

$22300\,kg \cdot \frac{1\,L}{0.803\,kg} = 27{,}770.8593$

$-7{,}682.\ I$

$20{,}088.859$

Round

$20{,}088.86\,L$

$10224.9\,L$

2.

$10000000\ lb \cdot \frac{4.5\,N}{1\,lb}$

$45{,}000{,}000\,N - 2{,}222{,}222.22 = 42{,}777{,}777.78$

$10{,}000{,}000\,N \cdot \frac{1\,lb}{4.5} = 2{,}222{,}222.22$

3. $\frac{1\,in}{25.4} =$

$1.25\,in \cdot \frac{25.4\,mm}{1\,in} = 31.75\,mm$

$1.25\,cm \cdot \frac{10\,mm}{1\,cm} = 12.5\,mm$

$31.75\,mm$
$-12.5\,mm$
18.25

18.2 mm off

5. $\frac{320\,gal}{hr} \cdot \frac{1\,hr}{60\,min} = 5.33\,gal/min$

$100\,gal \div 5.33\,gal/min \approx 19\,min$

6. $\frac{63.55\,g}{mol}$

$\frac{8.96\,g}{cm^3}$

$\frac{8.96\,g}{cm^3} \cdot \frac{mol}{63.55\,g} = \frac{0.14099\,mol}{cm^3} \cdot \frac{6.022E23\,atoms}{1\,mol}$

84,904,988,198,269,074,464,988.198269074

Quiz 1.A2

need help

?

2700 − 2.7615 = 2.697 → ~~2.697 E3~~

Level 1 HW: 1.A.2

1. AVG.

6.145 cm
6.12 cm
+ 6.2 cm
18.465 cm
↓ round
18.5 cm
÷ 3
6.17 cm ✓

Volume cm

6.145 cm
6.12 cm
× 6.2 cm
233.16588 cm³
↓
230 cm³

Volume m

230 cm³ · 1 m³ / 1 000 000 cm³
2.3E−4 m^3

$\frac{1m}{100cm}$ ✓ → $\frac{1m^3}{100cm^3}$ ✗

$\frac{1m}{100cm} \cdot \frac{1m}{100cm} \cdot \frac{1m}{100cm} = \frac{1m^3}{1000\,000\,cm^3}$ ✓

2.

Trial	Time	Speed (m/s)	Acceleration (m/s²)	
1	3.11	7.651	a1	
2	3.07	7.519	a2	
3	2.91	7.724	a3	
Avg.	c	d	Avg. Accel (column)	b
			Avg. Accel (from Avgs.)	e

Accel

7.651 m/s − 3.11 s = 2.46 m/s²
7.519 m/s − 3.07 s = 2.45 m/s²
7.724 m/s − 2.91 s = 2.65 m/s²

Avg of Accelerations

2.46 m/s²
2.45 m/s²
+ 2.65 m/s²
7.56 m/s²
÷ 3
2.52 m/s²

3. Avg time

2.91 s
3.07
2.91
+ 3.11
12.03 s
÷ 4
3.008 s

Avg speed

7.689 m/s
7.519
7.724
+ 7.651
30.583 m/s
÷ 4
7.6458 m/s

Avg Accel

7.6458 m/s ÷ 3.008 s
2.542 m/s²

Avg time

3.11 s
3.07 s
+ 2.91 s
9.09 s
÷ 3
3.03 s

Avg. velocity/speed

7.651 m/s
7.519
+ 7.724
22.894 m/s
÷ 3
7.6313 m/s

Avg Accel. from Avg's

7.6313 m/s ÷ 3.03 s = 2.52 m/s²

Level 2 HW: 1.A

3. 756 | 37 2 | 0 8 3 | + 2 5
796.53
↓
797

366.3
× 0.0032
1.17216
↓
1.2

5.620
× π
17.6557507
17.66

4)

13.0 acres · $\frac{43560\,ft^2}{1\,acre}$ = 566280 ft² · $\frac{0.09290304\,m^2}{1\,ft^2}$ = 52609.13349 m² = 52600 m²

$\frac{1\,ft^2}{0.09290304\,m^2}$ or $\frac{1\,ft}{0.3048\,m}$

height = 481 ft · $\frac{0.3048\,m}{1\,ft}$ = 146.6088 m = 147 m

$V = \frac{b \cdot h}{3} = \frac{52600\,m^2 \cdot 147\,m}{3} = 2\,580\,000\,m^3 = 2.58 \times 10^6\,m^3$

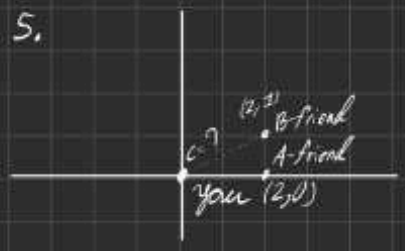
1. $kg\frac{m}{s^2} = k\frac{Qq}{r^2}$

$kg\frac{m}{s^2} = k\frac{C \cdot C}{m^2}$

$kg\frac{m^3}{s^2} = kC^2$

$k = kg\frac{m^3}{s^2 \cdot C^2}$

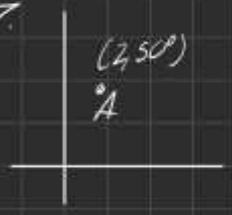
5.



$c^2 = a^2 + b^2$
$c^2 = 2^2 + 1^2$
$c^2 = 4 + 1$
$\sqrt{c^2} = \sqrt{5}$
$c = \sqrt{5}$

6. SOHCAHTOA
$\tan\theta = \frac{1}{2}$
$\theta = \tan^{-1}(\frac{1}{2})$
$\theta \approx 27^\circ$

7.



$B'(5, -50°)$

$x = r \cdot \cos(\theta)$
$y = r \cdot \sin(\theta)$

$x_1 = 2\cos(50) = 1.286$
$y_1 = 2\sin(50) = 1.532$

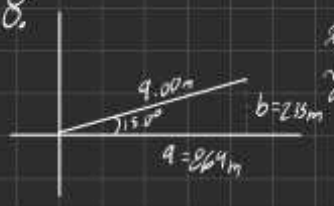
$x_2 = 5\cos(-50) = 3.214$
$y_2 = 5\sin(-50) = 3.830$

$D = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
$= \sqrt{(3214 - 1286)^2 + (-3830 - 1532)^2}$
$\approx 5.70\,m$

$\vec{V_1} - \vec{V_2} = \text{distance}$

$=$

8.



$x = 9 \cdot \cos(15) = 8.69$
$y = 9\sin(15) = 2.33$

9.

$a = 3.00 \cdot \sin(60.0°) = 2.60$
$b = 3.00 \cdot \cos(60.0°) = 1.50$

10. $\frac{1\,mi}{hr} \cdot \frac{1.61\,km}{1\,mi}$

$\frac{1.609344\,km}{1\,mi} \approx \frac{1.61\,km}{1\,mi}$

$\frac{55\,mi}{hr} \cdot \frac{1.61\,km}{1\,mi} = 88\,km/hr$

$\frac{65\,mi}{hr} \cdot \frac{1.61\,km}{1\,mi} \approx 100\,km/hr$

$100 - 88 = 12\,km/hr$

11. guess 700 years

$\$10{,}000{,}000{,}000{,}000 \cdot \frac{1\,sec}{1{,}000\,\$} = 1E10\,sec$

$1E10\,sec \cdot \frac{1\,min}{60\,sec} \cdot \frac{1\,hr}{60\,min} \cdot \frac{1\,day}{24\,hr} \cdot \frac{1\,year}{365\,day} \approx 300\,years$

$C = 2\pi r$
$= 2\pi\ 6378\,km$
$C = 40074.15589\,km$

$10\,000\,000\,000\,000 \cdot 15.5\,cm = 1.55E14\,cm \cdot \frac{1\,km}{100{,}000\,cm} = 1550000000\,km$

$1550000000\,km \div 40074\,km = 38700$ times

12. sec in year
$365\,day \cdot \frac{24\,hr}{1\,day} \cdot \frac{60\,min}{1\,hr} \cdot \frac{60\,sec}{1\,min} = 31536000\,sec$
$\downarrow$
$3.15E7$

$1\,m^3$

$1\,m^3/sec \cdot \frac{1}{10^{-6}\,m^3} = 1E18\,sec \cdot \frac{1\,min}{60\,sec} \cdot \frac{1\,hr}{60\,min} \cdot \frac{1\,day}{24\,hr} \cdot \frac{1\,year}{365\,day} = 3E10$

$\frac{1}{\sin 30} = \frac{x}{\sin 90}$
$2 = x$

$2^2 = a^2 + 1^2$
$4 - 1 = a^2$
$\sqrt{3} = a$

$A = uv \sin\theta$

$\begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ 3 & 0 & 0 \\ 3^{1/2} & 1 & 0 \end{vmatrix}$ $0\hat{x} - 0\hat{y} + (3 - 0)\hat{z} = [0, 0, 3]$

$3.0\,m^2$

~~[√3 m, 1.0 m]~~

[1.7 m, 1.0 m]

[2.0 m, 30 deg]

Vector Area

$V = \vec{A} \cdot \vec{L} = \vec{u} \times \vec{v} \cdot \vec{w}$ $\quad [1.0m, 0, \underline{3.0m}]$

$$\begin{bmatrix} \hat{x} & \hat{y} & \hat{z} \\ 3 & 0 & 0 \\ 3^{1/2} & 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} 1.0m \\ 0 \\ 3.0m \end{bmatrix} = \begin{vmatrix} 1.0 & 0 & 3.0 \\ 3 & 0 & 0 \\ 3^{1/2} & 1 & 0 \end{vmatrix} = 1(0-0) - 0(0-0) + 3.0(3-0) = 9.0m^3$$

2nd way

$B\cos\delta = z\hat{z}$

$\vec{A} \cdot \vec{B} = A\hat{z} \cdot B\cos\hat{z} = AB\cos\delta$

$3.0m \cdot 3.0m$

$9.0m^3$

Level 1 HW: 1.B2

Vector conversion

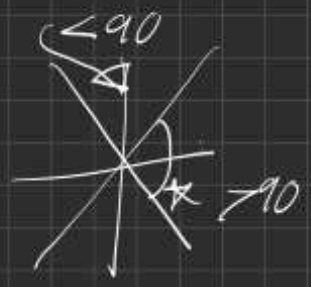
$V_1 = [185.81, 52.34°]$

$V_2 = [77.80, -46.07°]$

SOHCAHTOA

$x = r \cdot \cos(\theta)$

$y = r \cdot \sin(\theta)$

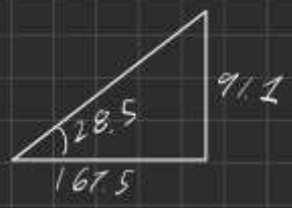


$x_1 = 185.81 \cdot \cos(52.34°)$

$y_1 = 185.81 \cdot \sin(52.34°)$ $\rightarrow [113.53, 147.10]$

$x_2 = 77.80 \cdot \cos(-46.07°)$

$y_2 = 77.80 \cdot \sin(-46.07°)$ $\rightarrow [53.98, -56.03]$



$V_1 + V_2 = [113.5, 147.1] + [53.98, -56.03] = [167.5, 91.1]$

$V_3 = [190.7, 28.5]$

$d = \sqrt{(167.5)^2 + (91.1)^2}$

$d = 190.7$

$$\text{Area} = V_1 \times V_2 = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ 113.53 & 147.10 & 0 \\ 53.98 & -56.03 & 0 \end{vmatrix} = (0-0) - (0-0) + (-6361 - 7940) = -14301$$

$$\text{Volume} \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ 113.53 & 147.10 & 0 \\ 53.98 & -56.03 & 0 \end{vmatrix} \begin{bmatrix} \quad \end{bmatrix}$$

Sig figs

$\text{Volume} = A \cdot \vec{w}$

$= |-14301| \cdot |135.21| \cdot \cos(38.26°)\hat{n}$

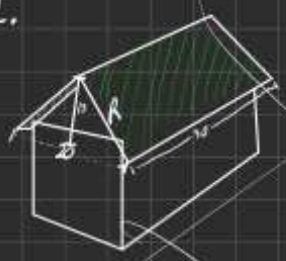
$= 1933600 \cdot 0.7852$

$= 1518000$

$= 1.518 \times 10^6$

Level 2 HW: Section 1.13 SOHCAHTOA

1.



$\vec{L}_R = [0, \sqrt{20^2+15^2}\,ft, \tan^{-1}\frac{15}{10}]$

$\vec{L}_R = [0, 25\,ft, 56\,deg]$

Convert to cartesian

$\vec{L}_R = [0, 20\,ft, 15\,ft]$

Area = Magnitude

Area of $\vec{L}_R$ & $\vec{L}_l \rightarrow L_R \times L_l$

$$[0,20,15]\times[30,0,0] = \begin{bmatrix} \hat{x} & \hat{y} & \hat{z} \\ 0 & 20 & 15 \\ 30 & 0 & 0 \end{bmatrix} = \hat{x}(0-0) - \hat{y}(0-450) + \hat{z}(0-600) = 450\hat{y} - 600\hat{z}$$

Area Vector $= [0, 450, -600]$

Area Vector $[0, 450, -600] \rightarrow$ polar

$\vec{V} = [x, y, z]$

$r = \sqrt{x^2+y^2+z^2}$

$\phi = \cos^{-1}\left(\frac{z}{r}\right)$

$\theta = \tan^{-1}\left(\frac{y}{x}\right)$

$\vec{V} = [r, \theta, \phi]$

$r = \sqrt{0^2+450^2+(-600)^2}$

$r = 750$

$\phi = \cos^{-1}\left(\frac{-600}{750}\right)$

$\phi = 143\,deg$

$\theta = \tan^{-1}\left(\frac{450}{0}\right)$

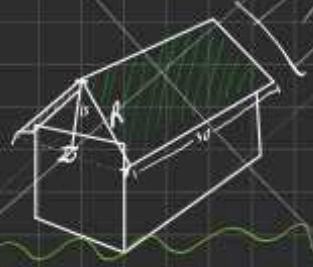
$\theta \approx 0\,deg$

Polar Area Vector $= [750\,ft^2, 0\,deg, 143\,deg]$

latitude 35.6870°N

March 20-21

perpindicular

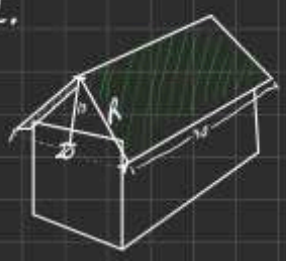


Level 2 HW: Section 1.13

SOHCAHTOA

Attempt #2

1.



$\vec{R} = [0, 10\,ft, 15\,ft]$

$\vec{L}_R = [0, \sqrt{10^2+15^2}\,ft, \tan^{-1}\frac{15}{10}]$

$\vec{L}_R = [0, 18\,ft, 56\,deg]$

Spherical

$[r, \phi, \theta]$

Convert to cartesian

$\vec{L}_R = [0, 10\,ft, 15\,ft]$

Area = Magnitude

Area of $\vec{L}_R$ & $\vec{L}_l \rightarrow L_R \times L_l$

$\phi \leftarrow$ Phi

pronounced "fee"

$[0, 10, 15] \times [30, 0, 0] = \begin{bmatrix} \hat{x} & \hat{y} & \hat{z} \\ 0 & 10 & 15 \\ 30 & 0 & 0 \end{bmatrix} = \hat{x}(0-0) - \hat{y}(0-450) + \hat{z}(0-300) = 450\hat{y} - 300\hat{z}$

Area Vector $= [0, 450, -300]$

$ft^2 \quad ft^2$

Area Vector $[0, 450, -300] \rightarrow$ polar

$\vec{V} = [x, y, z]$

$r = \sqrt{x^2+y^2+z^2}$

$\phi = \cos^{-1}\left(\frac{z}{r}\right)$

$\theta = \tan^{-1}\left(\frac{y}{x}\right)$

$\vec{V} = [r, \theta, \phi]$

$r = \sqrt{450^2 + (-300)^2}$

$r = 540\,ft^2$

$\phi = \cos^{-1}\left(\frac{-300}{541}\right)$

$\phi \approx 120\,deg$

$\theta = \tan^{-1}\left(\frac{450}{0}\right) \rightarrow$ Geometry

$\theta \approx 90°\,deg$

spherical Area Vector $= [540\,ft^2, 90°, 120°]$

1 panel $= 4ft \cdot 6ft$
$= 24\,ft^2$

$A = 540\,ft^2 \div 24$
≈ 22

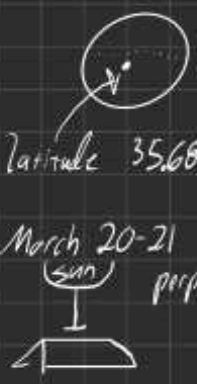
22 panels

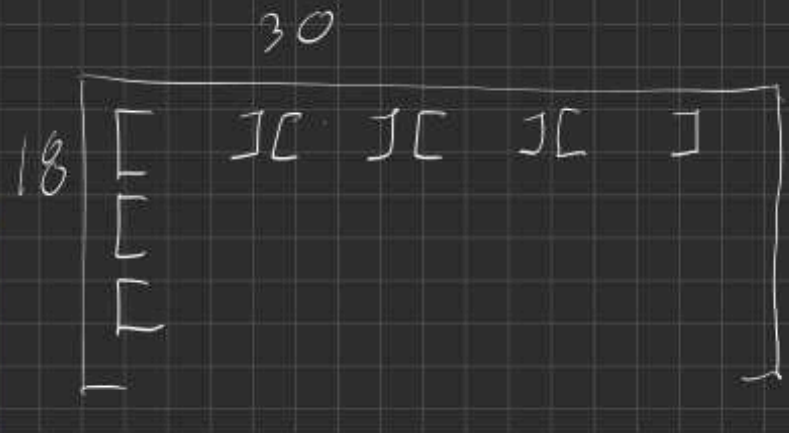
4x3

12 panels

latitude 35.6870°N

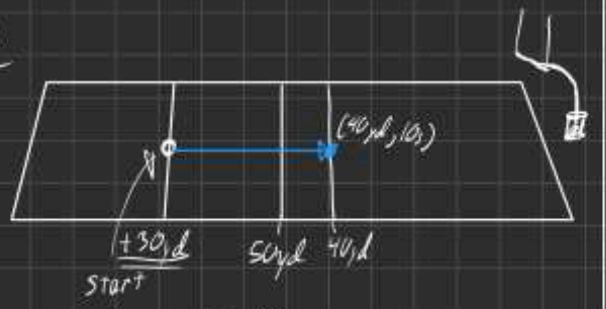
March 20-21





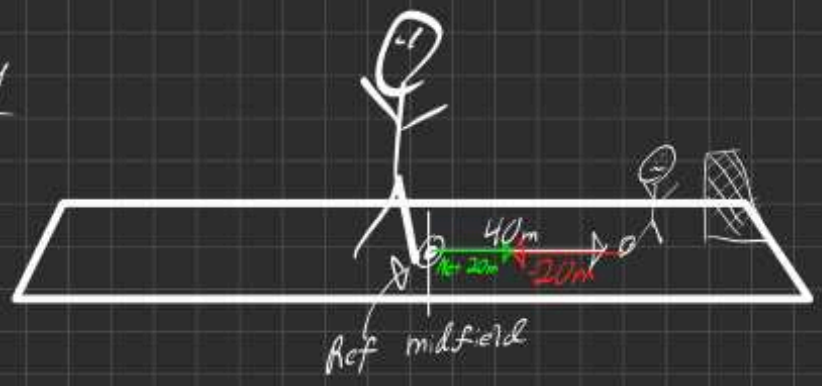
Vector Kinematics

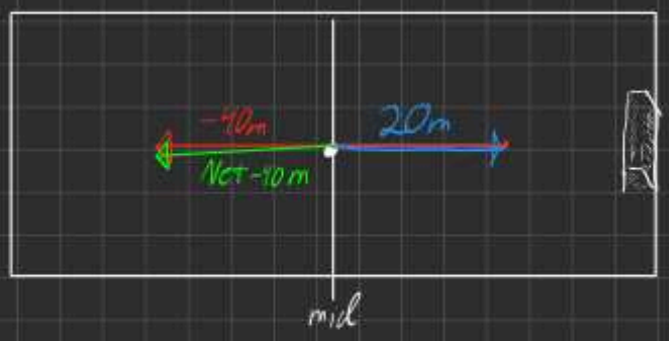
2



$$\text{Velocity} = \frac{\Delta y}{\Delta t} = \frac{30\,yd}{10\,s} = 3\,m/s$$

1

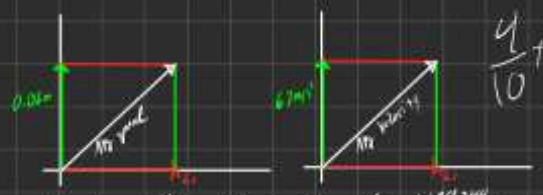




3.

0mph → 30mph in 2 s

30mph → 0mph in 2 s



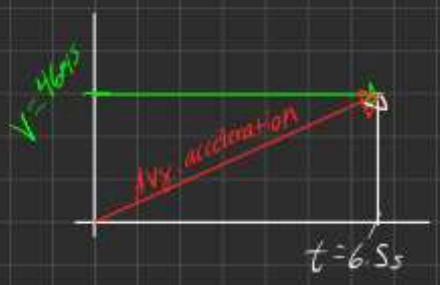
$\frac{4}{10}$ths

$$\text{Accel.} = \frac{\Delta V}{\Delta t} = \frac{30\,mph}{2\,s} \cdot \frac{1\,hr}{60\,min} \cdot \frac{1\,min}{60\,s} \cdot \frac{1609.34\,m}{1\,mi} \approx 6\ m/s^2$$

$$\text{Accel.} = \frac{\Delta V}{\Delta t} = \frac{-30\,mph}{2\,s} \cdot \frac{1\,hr}{60\,min} \cdot \frac{1\,min}{60\,s} \cdot \frac{1609.34\,m}{1\,mi} \approx -6\ m/s^2$$

Custom Problem

Let's say you want to calculate the Avg. acceleration of your hobby rocket. For the C6-5 rocket that you have chosen. It takes 6.5 seconds to reach its apogee(Highest point) which is 3.0*10^2m. Calculate the Avg. Velocity and Avg. Acceleration.



$$V = \frac{\Delta x}{\Delta t} = \frac{300\,m}{6.5\,s} = 46\,m/s \qquad a = \frac{\Delta V}{\Delta t} = \frac{46\,m/s}{6.5\,s} = 7.1\,m/s^2$$

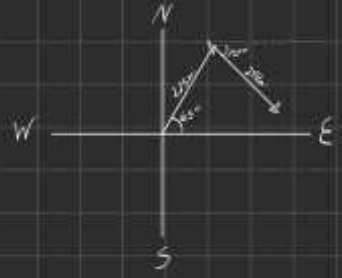
$$\boxed{\begin{array}{l} \text{Avg. } V = 46\,m/s \\ \text{Avg. } a = 7.1\,m/s^2 \end{array}}$$

$\hat{\theta}$ – Polar Cylindrical or Spherical azimuthal vector

$\hat{\rho}$ – spherical radial vector ϕ – Spherical polar vector

$\hat{r}$ – polar or Cylindrical radial vector

Vector displacement in 2D



$r_1 = (275\,mi, 65°)$ $\quad x = r\cos\theta \quad y = r\sin\theta$

$r_1 = (116mi, 249mi)$

$r_2 = (240\,mi, -70°)$

$r_2 = (82.1mi, -226mi)$

$\Delta r = r_1 + r_2 = (116mi, 249mi) + (82.1mi, -226mi) = \boxed{(198\ mi, 23.0mi)}$

$\Delta r \rightarrow$ polar $\quad \Delta r = (199mi, 6.63°)$

$x = r\cos(\theta)$

$\cos^{-1}\left(\frac{x}{r}\right) = \cos^{-1}(\cos(\theta))$

$\cos^{-1}\left(\frac{198}{199.33138}\right)$

$\theta \approx 6.63°$

$r = \sqrt{(198)^2 + (23)^2}$

$r = 199mi$

Level 1 HW 11.A.1

$speed = \frac{distance}{time} = \frac{200\,yd}{25\,s} = 8\,yd/s$

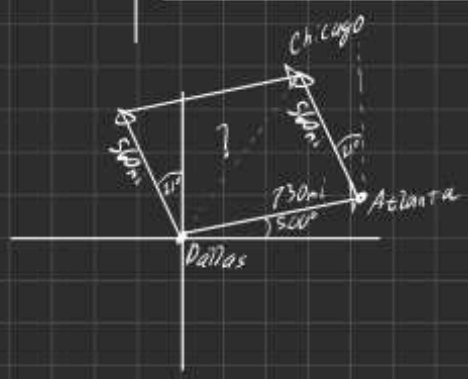
$V = \frac{\Delta x}{\Delta t} = \frac{displacement}{time} = \frac{100\,yd}{12.5\,s} = 8\,yd/s$

0.30mi across

tort = 0.60mph

hare = 0.30 mph for 0.30mi then 0.90mph

$V = \frac{\Delta x}{\Delta t} = \frac{0.60mi}{1hr}$



$x_1 = r\cos\theta$
$= 730\cos(5.00)$
$x_1 = 727mi$

$x_2 = r\cos\theta$
$= 560\cos(111)$
$x_2 = -201mi$

$y_1 = r\sin\theta$
$= 730\sin(5)$
$y_1 = 63.6mi$

$y_2 = r\sin(\theta)$
$= 560\sin(111)$
$y_2 = 523mi$

$\vec{V}_1 = [727mi, 63.6mi]$; $\vec{V} = [-201mi, 523mi]$

$\vec{V}_1 = [727mi, 63.6mi] + [-201mi, 523mi] = [526mi, 587mi]$

$r = \sqrt{(526)^2 + (587)^2}$

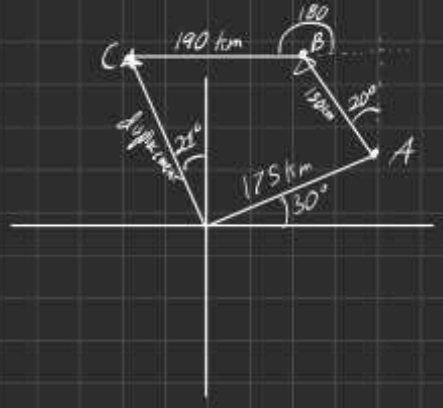
$r \approx 788mi$

$\theta = \tan^{-1}\left(\frac{587}{526}\right)$

$\theta \approx 48°$

$\vec{r} = [788mi, 48°]$

4.



$V_1 = [175km, 30.0°] = [152, 87.5]$

$V_2 = [150km, 110°] = [-51.3, 141]$

$V_3 = [190km, 180°] = [-190, 0]$

SOHCAHTOA

$V_1 + V_2 = [152, 87.5] + [-51.3, 141] = [101, 228.5] + \overset{V_3}{[-190, 0]} = [-89, 228.5]$

$V_4 = [-89, 228.5]$

$r = \sqrt{(-89)^2 + (228.5)^2}$

$r \approx 250$ km

$\theta = \tan^{-1}\left(\frac{228.5}{|-89|}\right)$

$\theta \approx 180 - (68.7)$

$\theta \approx 111$

3 sig figs

$a = \frac{\Delta V}{t}$

9. $\frac{1mi}{1609.344m}$

$\frac{3 \times 10^8 m}{s} \cdot \frac{1mi}{1609.344m} \cdot 0.1s = 2mi$

Circumference $= \pi d$

$\pi \cdot 7926$

$C = 24900mi$

$24900mi = 8E^{-5}$

10. $V_i = \frac{3.75m}{0.500s} = 7.5m/s$ Avg $= \frac{7.5 + 0.894}{2} = 4.197$

$a = \frac{\Delta V}{\Delta t} = \frac{7.5 - 0.894}{1} = 6.61 m/s^2$

11. $V_x(t) = 60.m/s + (0.50m/s^3)t^2$

Jerk

ΔV for $t = 1.8$ $t = 3$

$V(1) = 60.m/s + (0.50m/s^3)(1s^2)$

$= 60.50 m/s$

$V(3) = 60m/s + (0.50m/s^3)(3s)^2$

≈ 64.5

$\Delta V = 64.5 - 60.50 = 4 m/s$

$a = a_0$

$V_f = a(t) + V_0$

$x = \frac{1}{2}at^2 + V_0(t) + x_0$

Level 1 HW: 11.A.2

1. $\vec{F} = m\vec{a}$

$\vec{F} = 6.0\,kg \cdot 2.0\,m/s^2$

$\vec{F} = 12\,\frac{kg\,m}{s^2} \rightarrow 12N$

$\vec{F} = m\vec{a}$

$(12N = 4.0\,kg \cdot \vec{a}) \div 4.0\,kg$

$3\,m/s^2 = \vec{a}$

2. $\vec{F} = m\frac{\Delta V}{\Delta t}$

$7.5\times10^8 N = 1.5\times10^7\,kg \cdot \frac{\Delta V}{\Delta t}$

$0.05\,m/s^2 = \frac{\Delta V}{\Delta t}$

$\Delta V = 80\,km/h - 0$

$\Delta t = \frac{22\,m/s}{0.05\,m/s^2}$

$\Delta V = \frac{80\,km}{h} \cdot \frac{1\,h}{60\,min} \cdot \frac{1\,min}{60\,sec} \cdot \frac{1000\,m}{1\,km} = 22\,m/s$

$\Delta t = 440\,s$

3. $\vec{F} = m\frac{\Delta V}{\Delta t}$

$\Delta V = 320\,m/s - 0$

$\Delta V = 320\,m/s \cdot \frac{1\,km}{1000\,m} = 0.32\,km/s$

$\vec{F} = 5.0\,g \cdot \frac{0.32\,km/s}{0.0026\,s}$

$\vec{F} = 5.0\,g \cdot 120\,km/s^2$

$\vec{F} = 620N$

$\frac{320\,m}{s} = \frac{0.82\,m}{t}$

$t\,320\,m = 0.82\,m/s$

$t = 0.82\,m / 320\,m \cdot s$

$t = 0.0026\,s$

$d = \frac{\Delta V}{2\Delta t}$

4. Force for 3G on 70.0 kg

$3G = 3(9.81)\,m/s^2$

$N = 9.81\,m/s^2 \cdot x\,kg$

$N = 29.4\,m/s^2 \cdot 70.0\,kg$

$N = 2.06\times10^3 N$

$4.448N = 1\,lb$

$2058N \cdot \frac{1\,lb}{4.448N} = 500\,lb$

5.

$\vec{F} = m\vec{a}$

$200N = 1000\,kg \cdot \vec{a}$

$0.2\,m/s^2 = \vec{a}$

$V = at$

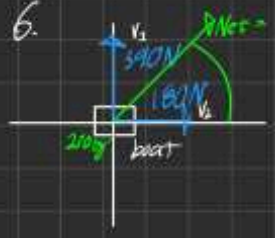
$V = 0.2\,m/s^2 \cdot 10\,s$

$V = 2\,m/s$

$\frac{0.2\,m}{s^2} = \frac{x\,m}{(10\,s)^2}$

$20\,m = x\,m$

6.



$V_1 + V_2 = [0, 390N] + [180N, 0] = [180N, 390N]$

$[180N, 390N] =$

$r = \sqrt{(180)^2 + (390)^2}$

$r = \sqrt{32400 + 152100}$

$r = 430\,N$

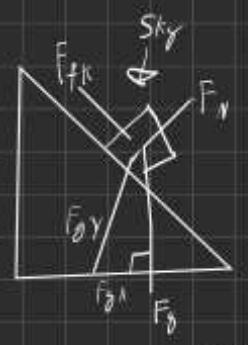
$\theta = \tan^{-1}\left(\frac{390}{180}\right)$

$\theta = 65°$

$V_3 = [430N, 65°]$

$f = \mu N$

$\sum F_x = F_x - f_x$



$\sum F_y = F_N - F_{gy} = 0N$

$\sin 30 = \frac{F_{gx}}{5(9.8)}$

$\sum F_x = F_{gx} - F_{fk} = 5kg(a)$

$\cos 30° \frac{F_{gy}}{5(9.8)}$

$F_{gx} = 24.5N$

2.3

$F_{gy} = 42.44N$

$\sum F_x = \sum F_{gx} - F_{fs} = 0N$

Bi Weekly 3 Corrections

Newtons 1st Law: A body remains at rest, or in motion unless acted upon by a force.

A man steps off a ledge, and falls 3.0m. How long does the fall take.

$\sum F = -mg = m(-g)$

$\Delta y = -\frac{1}{2}gt^2 + \cancel{V_0 t}$

$3 = -\frac{1}{2}(9.81)(t)^2$

$3 = -4.905(t)^2$

$\sqrt{-0.6116} = \sqrt{t^2}$

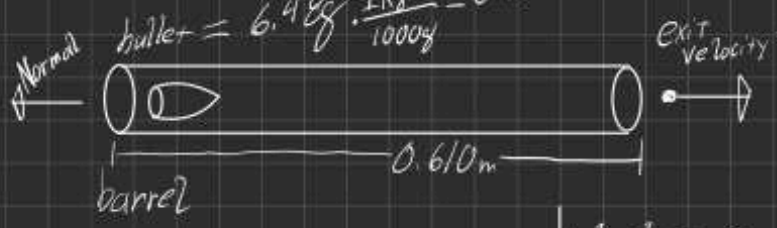
$0.782 = t$

$\boxed{t \approx 0.78s}$

HW. Level 1: 11.A.3

gun

bullet $= 6.48g \cdot \frac{1kg}{1000g} = 0.00648kg$



$V_f^2 = V_i^2 + 2ad$

$2ad = V_f^2 - V_i^2$

$a = \frac{V_f^2 - V_i^2}{2d}$

$a = \frac{(902m/s)^2}{2 \cdot 0.610m} = \frac{813604 m^2/s^2}{1.22m} = 6.67 \times 10^5 m/s^2$

$F = ma$

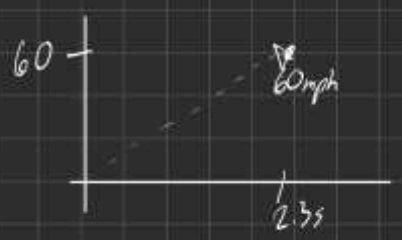
$F = 0.00648kg \cdot 6.67 \times 10^5 m/s^2$

$F = 4.32 \times 10^3 N$

$4320 N \cdot \frac{1lb}{4.448N} = 971 lbs$

4 "yes"

Muscle Cars



$\frac{60mi}{hr} \cdot \frac{1609.344m}{1mi} \cdot \frac{1hr}{60min} \cdot \frac{1min}{60sec} = 26.82 m/s$

$a = \frac{\Delta V}{\Delta t}$

$a = \frac{26.82 m/s}{2.3s} = 12 m/s^2$

top speed $= \frac{261mi}{h} \cdot \frac{1609.344m}{1mi} \cdot \frac{1hr}{60min} \cdot \frac{1min}{60s} = 116.68 m/s = V$

$a = \frac{\Delta V}{\Delta t} \rightarrow \Delta t = \frac{\Delta V}{a}$

$\Delta t = \frac{116.68 m/s}{12 m/s^2}$

$\Delta t = 10s$

$V_f^2 = V_i^2 + 2ad$

$d = \frac{V_f^2}{2a}$

$= \frac{(116.68 m/s)^2}{2 \cdot 12 m/s^2}$

$= \frac{13614 m^2/s^2}{24 m/s^2} = 567m \rightarrow 570m?$

(1 football field 91.44m)

$\rho_{air} = 1.22 kg/m^3$ $C_k = 0.39$ $A = 2.07m^2$

$\frac{250mi}{hr} \cdot \frac{1609.344m}{1mi} \cdot \frac{1hr}{60min} \cdot \frac{1min}{60s} = 111.76 m/s$

$\frac{250mi}{hr} \cdot \frac{1609.344m}{1mi} \cdot \frac{1hr}{3600s}$

$V = 112 m/s$

$V^2 = 12500 m^2/s^2$

$D = \frac{1}{2} \rho A C V^2$

$D = \frac{1}{2} \cdot 1.22 kg/m^3 \cdot 2.07m^2 \cdot 0.39 \cdot 12500 m^2/s^2$

$D = 6160N \rightarrow 6.16 \times 10^3 N \rightarrow$ accel

$F = ma$

$a = F/m$

$a = 6.16 \times 10^3 N \div 1900 kg$

$a = 3.24 m/s^2$

$\frac{1 ton}{736 kW}$

$W = F \cdot V$

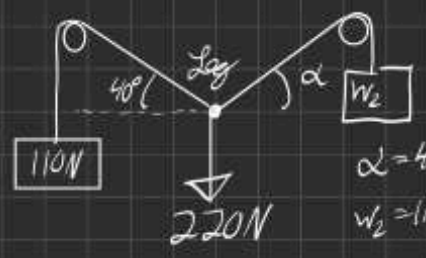
$W = 6160N \cdot 112 m/s$

$W = 690000 \frac{kg \cdot m^2}{s^3}$

$W = 690 \times 10^3 W \rightarrow W = 690. kW$

$\frac{W_D}{W_{tot}} \cdot 100 = \%$ for power

$\frac{690. kW}{736 kW} \cdot 100 = 93.8\%$ of an engine power is used to overcome drag at this velocity!!!



$\alpha = 40°$

$w_2 = 110N$

$$\sum F = (-\cos(40)\cdot 110N + \cos\alpha \cdot w_2)\hat{x} + (\sin(40)\cdot 110N + \sin\alpha \cdot w_2 - 220N)\hat{y}$$

$$\sum F_y = \sin(40)\cdot 110N + \sin\alpha \cdot w_2 - 220N = 0$$

$$\sin\alpha\, w_2 = -\sin(40)\cdot 110 + 220N$$

$$\sin\alpha\, w_2 = 149.29N$$

$$\sum F_x = -\cos(40)\cdot 110N + \cos\alpha \cdot w_2 = 0$$

$$\cos\alpha \cdot w_2 = \cos(40)\cdot 110N$$

$$\cos\alpha \cdot w_2 = 84.26N$$

$$\frac{\sin\alpha\, w_2}{\cos\alpha\, w_2} = \frac{149.29N}{84.26N}$$

$$\tan\alpha = 1.77$$

$$\alpha = \tan^{-1}(1.77)$$

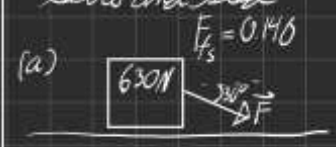
$$\alpha = 60.5°$$

$$\cos(60.53)\cdot w_2 = 84.26N$$

$$w_2 = 171N$$

Sled and slab

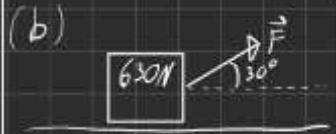
(a)



$$\sum F = mg - N + \tan(\theta)F + N_\mu$$

$$= (F\cos(\theta) - F_f)\hat{x} + (N - mg + F\sin\theta)\hat{y}$$

(b)



56N ← Net 13N →

Case (a) — $\boxed{\mu_k = 0.100 \quad F = 120N}$ — Case (b)

$$(F\cos(\theta) - F_f)\hat{x} + (N - mg + F\sin\theta)\hat{y}$$

11

(a)

$$y\begin{cases} F\sin(30°) + mg = N_{normal} \\ 120N\cdot\sin(30°) + 630N = N_{normal} \\ 690 = N_{normal} \end{cases}$$

$$-x\begin{cases} F_f = N\cdot\mu_k \\ = 690N\cdot 0.100 \\ F_f = 69N \end{cases}$$

$$+x\begin{cases} \sum F = F\cdot\cos(\theta) - F_f \\ = 120N\cdot\cos(30°) - 69N \\ = 103.92 - 69N \\ F = 34.9N \end{cases}$$

!!! $F = ma \qquad m = \frac{N}{9.81} = 64.2kg$

$$a = \frac{F}{m}$$

$$a = \frac{34.9N}{64.2kg} = 0.54 m/s^2$$

(b)

$$-y\begin{cases} \sum F = mg \\ = 630N \end{cases}$$

$$+y\begin{cases} \sum F_{Norm} = (F\cdot(-\sin(30)) + N) \\ = (120N\cdot(-\sin(30°)) + 630N) \\ \sum F_{Norm} = 570N \end{cases}$$

$$-x\begin{cases} F_f = N\cdot\mu_k \\ = 570N\cdot 0.100 \\ F_f = 57N \end{cases}$$

$$+x\begin{cases} \sum F = F\cdot\cos(30°) - F_f \\ = 120N\cdot\cos(30°) - 57N \\ \sum F = 46.9N \end{cases}$$

$F = ma \qquad m = 64.2kg$

$$a = \frac{F}{m} = \frac{46.9N}{64.2kg} = 0.73 m/s^2$$

(a)

$V_f^2 = V^2 + 2ad$

$V_f^2 = 0 + 2ad$

$V_f = \sqrt{2ad}$

$V_f = \sqrt{2 \cdot 0.54 m/s^2 \cdot 20m}$

$V_f = \sqrt{2.16 m^2/s^2}$

$V_f = 1.47 m/s$

(b)

$V_f = \sqrt{2 \cdot 0.73 m/s^2 \cdot 20m}$

$V_f = 1.71 m/s$

force needed for a const. V?

(a)

$69N = F_f$

(b)

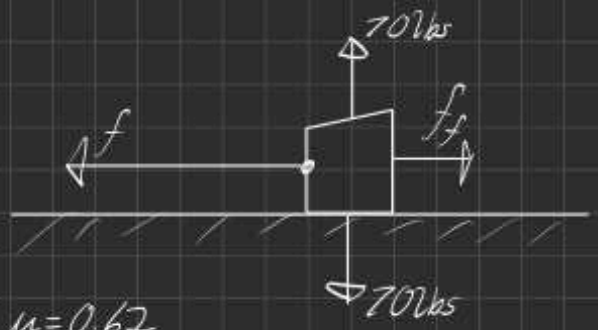
$57N = F_f$

Level 2 HW: Section 11.A

1. Max pull = 30% · body weight.

body weight = 210 lbs

Max = 0.30 · 210 lbs
= 63 lbs



$\mu = 0.62$

$F_f = \mu \cdot N$
$= 0.62 \cdot 70\,lbs$
$F_f = 43.4\,lbs \cdot \frac{4.4482\,N}{1\,lb} = 193\,N \approx 190\,N$

Cite: astroflow.com
Mars accel. from gravity

$g = G\frac{M}{r^2}$
$G \approx 6.674 \times 10^{-11}\,m^3\,kg^{-1}\,s^{-2}$
$M = 6.4171 \times 10^{23}\,kg$
$r = 3.3895 \times 10^6\,m$
$g = \frac{(6.674 \times 10^{-11}) \cdot (6.4171 \times 10^{23})}{(3.3895 \times 10^6)^2}$
$g = 3.71\,m/s^2$

Mars
$W = m \cdot g$
$W = 74.8\,kg \cdot (3.71\,m/s^2)$
$W = 278\,N \cdot \frac{0.2248\,lb}{1\,N} = 62.5\,lbs$

Earth
$W = 74.8\,kg \cdot 9.81\,m/s^2$
$W = 734\,N \cdot \frac{0.2248\,lb}{1\,N} = 165\,lbs$

difference
$165\,lbs - 62.5\,lbs = 103\,lbs$

---

3. Convert to military
1:55:58 → 13:55:58
2:45:17 → 14:45:17 −
1:29:59
89.59 sec

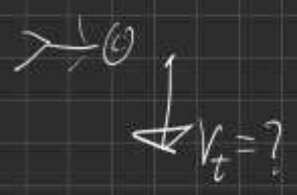
drop distance
24900 ft
− 7000 ft
17,900 ft · $\frac{0.3048}{1\,ft}$ = 546 m

$\Delta x = v_i t + \frac{1}{2} a t^2$
$\Delta x - v_i t = \frac{1}{2} a t^2$
$\frac{2\Delta x - v_i t}{t^2} = a$
$\frac{2 \cdot 17900\,ft - (0)}{(89.59\,s)^2} = a$
$\frac{35800\,ft}{8{,}026.36\,s^2} \approx 0.680\,m/s = a$

$F = ma$
$F = 90\,kg \cdot 0.680\,m/s$
$F = 61.2\,N$

762 m

$B = \rho_f V_{obj} \cdot g$
$612 = \frac{1}{2}(1.225 + 0.5452) \cdot V_{obj} \cdot 9.81$
$V_{obj} = \frac{612}{0.8851 \cdot 9.81}$
$V_{obj} = 7.05\,m^3 \cdot \frac{35.315\,ft^3}{1\,m^3} = 249\,ft^3$
≈ 242



$F_g = F_D$

$m \cdot g = \frac{1}{2} \rho V_t^2 A C_d$

$V_t = \sqrt{\frac{2 \cdot m \cdot g}{\rho \cdot A \cdot C_d}}$

$V = \sqrt{\frac{2 \cdot 90kg \cdot 9.81 m/s^2}{0.8851 \cdot 0.5m^2 \cdot 1.2}}$

$= \sqrt{\frac{1766 kg\, m/s^2}{0.5311 m^2}}$

$V_t \approx 57.66 m/s \cdot \frac{2.2369 mph}{1 m/s} = 129 mph$

$\Delta V = at$

$V_f = a t$

$t = \frac{V_f}{a}$

$t = \frac{57.66 ms}{0.68 m/s^2}$

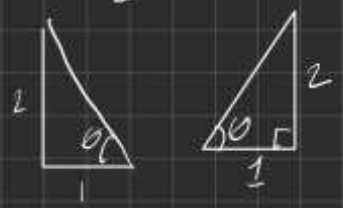
$t \approx 84.8 s$

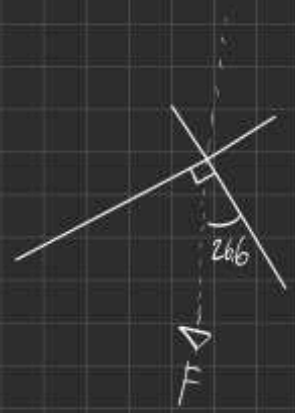
10. $\tau = r F \sin\theta$

$= 40.0 \cdot 6672 \cdot \sin(26.6)$

$= 1.19 \times 10^5 Nm \cdot \frac{0.7376}{1 Nm}$

$\tau = 8.81 \times 10^4 lb\text{-}f$





$\tan\theta = \frac{o}{a}$

$\theta = \tan^{-1}\left(\frac{2}{1}\right)$

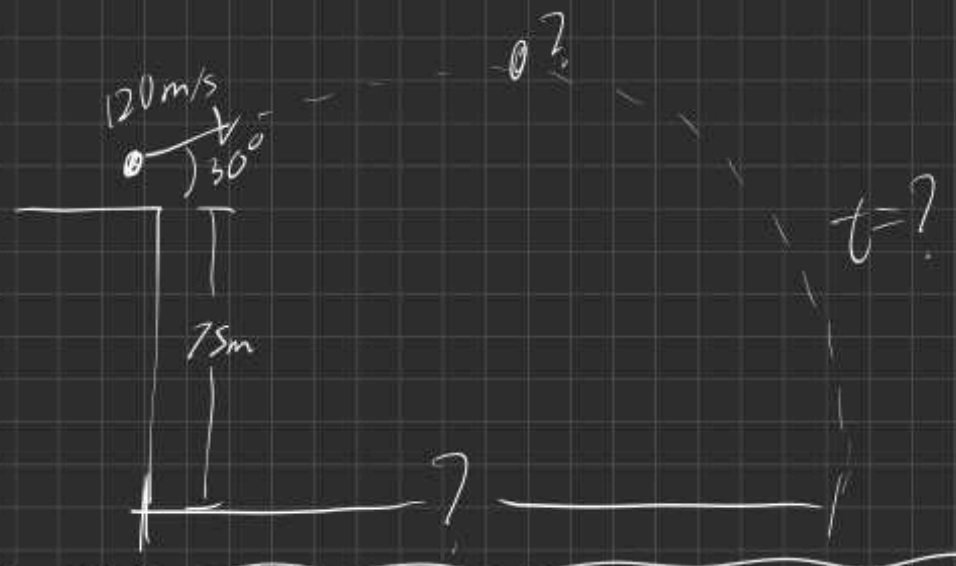
$\theta \approx 63.4°$

$\phi = 90 - 63.4°$

$\phi = 26.6°$

$r = 40m$

$1500 lb \cdot \frac{4.448}{1 lb} = 6{,}673 N$



$\Delta y = v_{iy} t + \frac{1}{2} a_y t^2$

$v_{iy} = 120 \sin 30 = 60 \, m/s$

$v_{ix} = 120 \cos 30 = 103.9$

$-75 = 60t + \frac{1}{2}(-9.81)t^2$

$0 = -4.905t^2 + 60t + 75$

$= \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ $\quad x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

$= \frac{-60 \pm \sqrt{60^2 - 4(-4.905)(75)}}{2(-4.905)}$

$\frac{-60 \pm 71.21}{-9.81} \approx 13.38 \, sec$

time

Range

$\Delta x = a \cdot t$

$\Delta x = 103.9 \cdot 13.4$

$\Delta x = 1392$ meters

Max height

$v_{fy}^2 = v_{iy}^2 + 2a_y \Delta y$

$0 = 60^2 + 2 \cdot -9.81 \cdot \Delta y$

$0 = 3600 - 19.62 \Delta y$

$-3600 = -19.62 \Delta y$

$+183.49 m = \Delta y$

$183.49 + 75 = 258 m$

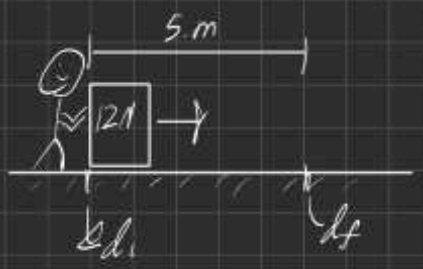
$$I = \int r^2 \, dm$$
$$= \int x^2 \, dm$$

$$\frac{dm}{m} = \frac{dx}{l}$$

$$I = \left(\frac{m_1 m_2}{m_1 + m_2}\right) l^2$$

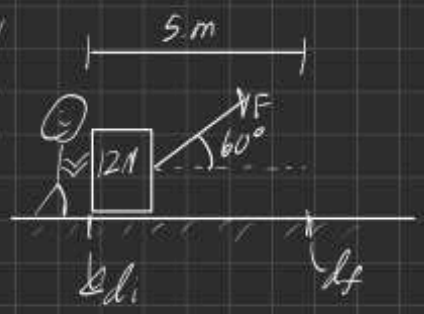
$$I = 17.78 \, l^2$$

Activity : Work



$$W = \int \vec{F} \cdot d\vec{r}$$

$$W = 12N \cdot 5.0m$$
$$W = 60Nm$$



$$W = \vec{F} \cdot d$$
$$W = 12 \cdot \cos(60) \cdot 5$$
$$W = 5 \cdot 6$$
$$W = 30. Nm$$

Derivatives

$$x(t) = \underset{\text{jerk}}{2.5t^3} + \underset{\text{accel}}{5.0t^2} + \underset{\text{vel}}{10t} + \underset{\text{const}}{15}$$

$$V(t) = 7.5t^2 + 10t + 10$$

$$a(t) = 15t + 10$$

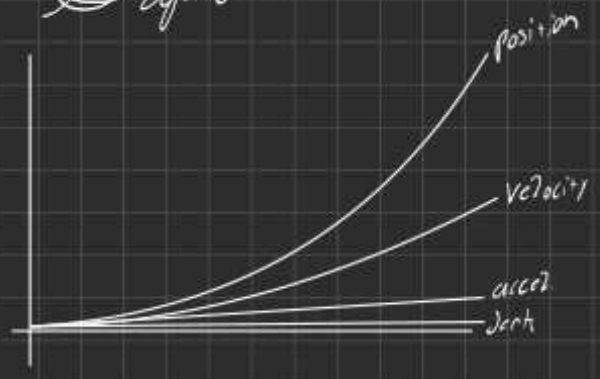
$$\text{Jerk}(t) = 15$$

Velocity at $t = 1$ is?

$$V(t) = 7.5t^2 + 10t + 10.$$

$$V(1) = 27.5 \, m/s$$

Graph



This is not realistic for a fly because the velocity of the fly is to high.

Level 1 HW: II.B.1.1 Linear Motion Practice

const. $a = ?$ $m = 1157kg$

$\Delta x = 240m$

$V_i = 0$ $V_f = \frac{120km}{hr} \cdot \frac{1hr}{60min} \cdot \frac{1min}{60sec} \cdot \frac{1000m}{1km} = 33m/s$

$\Delta V^2 = 2a\Delta x$

$33^2 = 2 \cdot a \cdot 240$

$\frac{1111}{480} = a$

$2.27 m/s^2 = a$

$\Delta V = at$

$33m/s = 2.27m/s^2 \cdot t$

$15sec = t$

$P = \frac{W}{\Delta t}$

$P = \frac{630240Nm}{15sec}$

$P = 42016Nm/s \cdot \frac{1hp}{745.7Nm/s} = 56hp$

$W = F \cdot \Delta d$

$W = 2626N \cdot 240m$

$W = 630240Nm$

$F = m \cdot a$

$F = 1157kg \cdot 2.27m/s^2$

$F = 2626kgm/s^2$

$F = 2626N$

Uniform acceleration

$V_i = 0$ $t_i = 0$

$V_f = 24.0m/s$ $t_f = 2.95s$

$a = ?$

$\Delta V^2 = 2a\Delta x$

$a = \frac{\Delta V^2}{2 \cdot \Delta x}$

$a = \frac{(24.0m/s)^2}{2 \cdot 35.4m}$

$a = 8.14m/s^2$

$\Delta x = \frac{1}{2}(\Delta V)t$

$= \frac{1}{2}(24.0m/s)2.95s$

$= 12m/s \cdot 2.95s$

$\Delta x = 35.4m$

Part 2

$V_i = 10.0m/s$

$V_f = 20.0m/s$

$a = 8.14m/s^2$

$V_f = V_i + at$

$\Delta V = at$

$\frac{\Delta V}{a} = t$

$\frac{10.0m/s}{8.14m/s^2} = t$

$1.23s = t$

$\Delta V = at$ $\Delta V = at$

$= 8.14 \cdot 1.23$ $= 8.14 \cdot 2.46$ $= 8.14 \cdot 4.92$

10 20 40

Will doubling the time always double the change in speed?

Yes

Because there is constant acceleration

A ball is launched vertically, from the ground, with a speed of 25.0m/s

How heigh does the ball go? 63.75m

How long does it take to reach the peak? 5.1s

How long does it take to hit the ground after peak? 7.9s

What is its velocity when it hits the ground?

w/ a mass of 148g, what is gravities work on the way up?

work done by gravity on the way down?

Net work done by gravity?

$t = \frac{2V_0 \sin\theta}{g}$

$t = \frac{2 \cdot 25.0m/s \cdot \sin(90°)}{9.81m/s^2}$

$t = \frac{50m/s}{9.81m/s^2}$

$t = 5.1s$

$\Delta y = -\frac{1}{2}gt^2 + \cancel{V_0 t}$

$t^2 = \frac{\Delta y}{\frac{1}{2} \cdot g}$

$\sqrt{t^2} = \sqrt{\frac{63.75}{4.905}}$

$t \approx 13s$

$\Delta y = \frac{1}{2}(V_i + V_f)t$

$= \frac{1}{2}(25.0m/s) \cdot 5.1s$

$= 12.5m/s \cdot 5.1s$

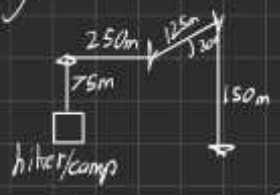
$\Delta y = 63.75m$

$\Delta V = at$

$\Delta V = 9.81m/s^2 \cdot 13s$

$\Delta V = 127.53m/s$

7)



$d_{tot} = \sum d$

$= 75 + 250 + 125 + 150$

$d_{tot} = \underline{600m}$

$position_f = \sum \vec{d}$

$= [0, 75] + [250, 0] + [108, 62.5] + [0, -125]$

$position_f = [358, 125]$

cart → polar

polar → cart

$[125, 30°]$

backward $= [-358, -125]$

$r = \sqrt{x^2 + y^2}$

$r = \sqrt{128320}$

$r \approx 358m$

$\tan\theta = \frac{y}{x}$

$\theta = \tan^{-1}\left(\frac{-125}{-358}\right)$

$\theta \approx 2°$

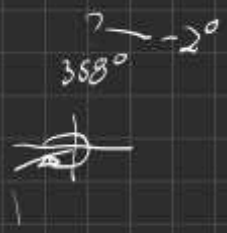
$[358m, 2°]$

$x = r\cos\theta = 125 \cdot \cos(30°) = 108m$

$y = r\sin\theta = 125 \cdot \sin(30°) = 62.5m$

$= [108m, 62.5m]$

vector to start $p_i = \underline{[-358m, -2°]}$



takes 15min ?avg mph?

$V = \frac{\Delta d}{\Delta t} = \frac{600m}{15min} \cdot \frac{1mi}{1609.34m} \cdot \frac{60min}{1hr} = \underline{1.49mph}$

reasonable? Yes

m/min = 40

$358m \cdot \frac{1min}{40m} = \underline{8.95min}$

Activity: Initial Velocity Comp.

Velocity vector cartesian components → ?

$\sin\theta = \frac{o}{h}$
$\sin\theta \cdot h = opp$
$\sin(30) \cdot 20 = opp$
$10 m/s = opp$

$\cos\theta = \frac{a}{h}$
$\cos(30)\, 20 = adj$
$17 m/s = adj$

[17_m/s, 10_m/s]

Start to derive all kinematic equations considering constant acceleration.

$$a = \frac{dv}{dt} = a_0 \quad \text{constant}$$

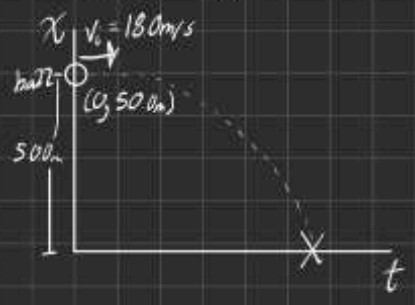
$$dv = a_0 \cdot dt$$

Quiz 1.B.2

Level 1 HW: II.B.1.2: Projectile motion

$$V = V_0 + at$$
$$\Delta x = V_0 t + \tfrac{1}{2}at^2$$
$$\Delta V^2 = 2a\Delta x$$
$$\Delta x = \tfrac{1}{2}(V_0 + V)t$$

1. $V_i = [18\ m/s, 0]$



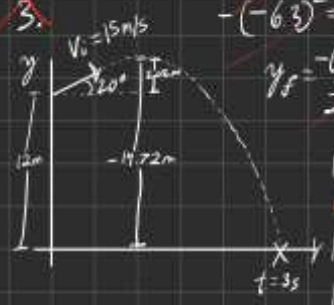
x v t

$$\Delta x = \tfrac{1}{2}(V_0 + V)t$$
$$\Delta y = \tfrac{1}{2}(V_0 + V)t$$
$$\Delta d = \tfrac{1}{4}(V_{0x} \cdot V_{0y} + V_x \cdot V_y)t^2$$

$$\Delta V^2 = 2 \cdot a \cdot \Delta y$$
$$V_f^2 - \cancel{V_i^2}^{0} = 2 \cdot a(y_f - y_i)$$
$$V_f^2 = -2 \cdot a \cdot y_i$$
$$V_f = \sqrt{-2(-9.81)(50)}$$

2. is wrong

3.



$$\Delta V^2 = 2a\Delta y$$
$$-(-6.3)^2 = 2(-9.81)y_f - \cancel{y_i}^{0}$$
$$y_f = \frac{-(-6.3)^2}{2(-9.81)} = 2.02m$$

$$\Delta V = aT$$
$$-(6.3 m/s) = (-9.81)t$$
$$t = 0.64s$$

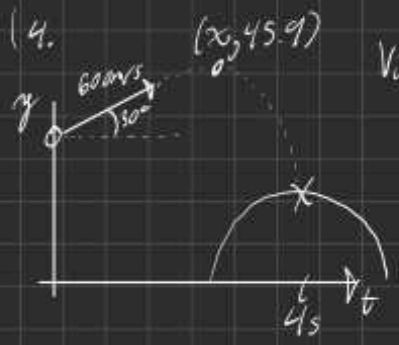
$$\Delta y = V_0 t + \tfrac{1}{2}at$$
$$y_f - y_i = V_0 t + \tfrac{1}{2}at$$
$$y_f = \tfrac{1}{2}(-9.81)(2.36)$$
$$y_f \approx 12m$$

$$t_u + t_d = t_t$$
$$t_d = t_t - t_u$$
$$t_d = 2.36s$$

$$y = -\tfrac{1}{2}gt^2 + V_0 \sin\theta t + \cancel{y_0}^{0}$$
$$y = -\tfrac{1}{2}(-9.81)(3)^2 + (15 \cdot \sin(20) \cdot 3$$
$$y = 25m$$

$= 60$ Or $6.5 \cdot 3$

4.



$V_i = [60, 30°]$

$$x = r\cos\theta = 520$$
$$y = r\sin\theta = 300$$
$[520, 30.0]$

$$\Delta V^2 = 2a\Delta y$$
$$\cancel{V_f^2}^{0} - V_i^2 = 2a(y_f - y_i)$$
$$-V_i^2 = 2ay_f$$
$$y_f = \frac{-(30)^2}{2(9.81)} = 45.9m$$

Activity: Angular Accelerations

$r = 0.300\,m$

ω = angular velocity

$\omega = \theta / t$

$\omega = 3600$ rev/min

V = tangential speed

$V = r\omega$

$3600 \frac{rev}{min} \cdot \frac{2\pi}{1 rev} \cdot \frac{1 min}{60 s} = 120\pi$ rad/s

$V = 0.300m \cdot 120\pi$ rad/s

$V = 113$ m/s

$a_c = \frac{v^2}{r}$

$= \frac{(113)^2}{0.300}$

$a_c = 4.26 \times 10^4$ m/s^2

$\alpha = \frac{\Delta\omega}{t} = \frac{-120\pi \text{ rad/s}}{1.67 s} = -226$ rad/s^2

$a_t = 0.300m(-226 \text{ rad/s}^2)$

$a_t = -67.8$ m/s^2

$a_{net} = \sqrt{a_c^2 + a_t^2}$

$a_{net} = \sqrt{(4.26 \times 10^4)^2 + (-67.8)^2}$

$a_{net} = 4.26 \times 10^4$ m/s^2

Activity: Angular Kinematics

$t = 1.67 s$

$\omega_f = 0$ rad/s

$\omega_i = 377$ rad/s

$\theta = \frac{1}{2}(\omega_i - \omega_f)t$

$\theta = \frac{1}{2} \cdot 377 \cdot 1.67$

$\theta = 315$ rad $\cdot \frac{1 rev}{2\pi rad} = 50.1$ rev

Module Quiz: 11.B.2

1. $V = r\omega$ ← $V_t = r \cdot \omega$

$\omega = \theta / t$

Level 1 HW: 1.C.2.2: Circular Kinematics

1. $r = 1\,ft$

distance of 60,000 mi = ? rads

$C = 2\pi r$
$= 2\pi \cdot 1$
$= 2\pi$
$= 6.28\,ft$

$60{,}000\,mi \cdot \frac{5280\,ft}{1\,mi} = 316{,}800{,}000\,ft \cdot \frac{1\,rev}{6.28\,ft} = 50420285.97 \approx 5.0\times10^7\,rev \cdot \frac{2\pi\,rad}{1\,rev} = 8.0\times10^6\,rad$

2. $\omega_i = 0$ $\quad \omega_f = 1.00\,rev/s$
$t = 30.0\,s$

$\alpha = \frac{d\omega}{dt} = \frac{1\,rev/s}{30.0\,s} = 0.0333\,rev/s^2 \cdot \frac{2\pi\,rad}{1\,rev} = 0.209\,rad/s^2$

$\times 2$

$d\omega = \alpha \cdot dt = 0.0666 \cdot 30.0\,s = 2.00\,rev/s$

3. $\omega_i = 0$ $\quad \omega_f = 2.51\times10^4\,rev/min \cdot \frac{1\,min}{60\,s} \cdot \frac{2\pi\,rad}{1\,rev} = 2.63\times10^3\,rad/s$
$t = 3.20\,s$

$\alpha = \frac{d\omega}{dt} = \frac{2.63\times10^3\,rad/s}{3.20\,s} = 8.21\times10^2\,rad/s^2 = \alpha$

$\Delta\theta = \frac{1}{2}(2.63\times10^3\,rad/s)\,3.20\,s$

$\Delta\theta = 4.21\times10^3\,rad$

4. $\omega_0 = 0$ $\quad \omega_1 = 5.0\,rev/s$ $\quad \omega_2 = 0$
$t_0 = 0$ $\quad t_1 = 8.0\,s$ $\quad t_2 = 20.0\,s$

1 $\quad$ 2

$\Delta\theta_1 = \frac{1}{2}(5)(8.0\,s)$ $\quad \Delta\theta_2 = \frac{1}{2}(5)\cdot 12$

$\Delta\theta_1 = 20\,rev$ $\quad \Delta\theta_2 = 30$

$\Delta\theta_1 + \Delta\theta_2 = \Delta\theta$
$\Delta\theta = 50\,rev$

5. $V_i = 29\,m/s$ $\quad V_f = 0$
$a = -1.75\,m/s^2$
$t = ?$

$\Delta x = V_i t + \frac{1}{2}at^2$

$\Delta V = at$
$\frac{-29\,m/s}{-1.75\,m/s^2} = t$
$t = 16.57\,s$

$x_f = 29 \cdot 16.57 + \frac{1}{2}(-1.75)(16.57)^2$
$x_f = 480.53 - 240.24$
$x_f = 240\,m$

$V_f^2 = V_i^2 + 2a\Delta x$
$V_f^2 = (29)^2 - 2(-1.75)(120.14)$
$V_f = \sqrt{841 - 420.49}$
$V_f = 20.5\,m/s$

$r = 0.330\,m$
$t = 16.6\,s$
$rev = ?$

$C = 2\pi r$
$= 2\pi \cdot 0.330$
$C = 2.07\,m/rev$

$240\,m \cdot \frac{1\,rev}{2.07\,m} = 116\,rev$

$V = r\omega$
$\omega = \frac{V}{r}$
$\omega = \frac{20.5}{0.33} = 62.1\,rad/s$

6. $\omega_i = 1.00\times10^2\,rev/min \cdot \frac{2\pi\,rad}{1\,rev} \cdot \frac{1\,min}{60\,s} = 10.5\,rad/s$
$\omega_f = 0$
$\alpha = -200\,rad/s^2$ $\quad t_f = ?$

$\Delta\omega = \alpha t$
$t = \frac{\Delta\omega}{\alpha}$
$t = \frac{-10.5\,rad/s}{(-200\,rad/s^2)}$
$t = 5.25\,s$

$rev = \frac{1}{2}(1.00\times10^2\,rev/min)\,0.0875\,min$
$rev = 50\,rev/min \cdot 0.0875\,min$
$rev = 4.38\,rev \cdot \frac{2\pi\,rad}{1\,rev} = 27.5\,rad$

7. diameter $= 10.0\,in \cdot \frac{0.0254\,m}{1\,in} = 0.254\,m$ $\quad r = 0.127\,m$

$\omega_i = 0 \quad \omega_f = 78.0\,rev/min \cdot (1/60) = 1.3\,rev/s \cdot \frac{2\pi\,rad}{1\,rev} = 8.17\,rad/s$

$t_i = 0 \quad t_f = 3.00\,s$

$$\alpha = \frac{d\omega}{dt} = \frac{1.3\,rev/s}{3.00\,s} = 0.433\,rev/s^2 \cdot \frac{2\pi\,rad}{1\,rev} = 2.72\,rad/s^2$$

$a_t = r\alpha$

$a_t = 0.127\,m \cdot 2.72\,rad/s^2$

$a_t = 0.345\,m/s^2$

even a 1 sec later

$a_t = 0.345\,m/s^2$

$v_t = r\omega$

$= 0.127\,m \cdot 8.17\,rad/s$

$v_t = 1.04\,m/s$

$$a_c = \frac{v^2}{r} = \frac{(0.345\,m/s)^2}{0.127\,m} = 0.937\,m/s^2$$

$a_{net} = \sqrt{a_c^2 + a_t^2}$

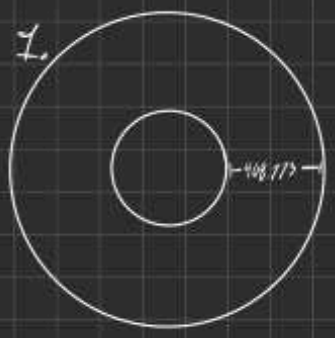
$= \sqrt{(0.937)^2 + (0.345)^2}$

$= \sqrt{0.997}$

$a_{net} \approx 0.998\,m/s^2$

Level 2 HW: Section 11.B

1.



A

$d = r \cdot t$

$t = \frac{d}{r}$

$t = \frac{408,773\,km}{350.\,km/h}$

$t = 1.17\,hr$

B

smaller

$\frac{19}{1.17} = 16.2$ times

C

D

$\Delta v = at$

$\frac{v_f}{a} = t$

$t = \frac{972\,m/s}{3.981\,m/s^2}$

$t = 3.3\,s$

$1\,hp = 745\,Watts$

$\frac{350.\,km}{hr} \cdot \frac{1\,hr}{60\,min} \cdot \frac{1\,min}{60\,s} \cdot \frac{1000\,m}{1\,km} = 97.2\,m/s$

E

$P = F v$

$= F_{hold} + F_{arm}$

$P = (981 + (1\,kg \cdot (3.981))) \cdot 97.2$

$P = 3814\,Nm$

$= 3814\,Watts \cdot \frac{1\,hp}{745\,Watts}$

$= 5.11\,hp$

E

2. $h = 41,425\,m$

$v_i = 0$

$v_f = \frac{822\,mi}{hr} \cdot \frac{1609.344\,m}{1\,mi} \cdot \frac{1\,hr}{60\,min} \cdot \frac{1\,min}{60\,s} = 367\,m/s$

$y_i = 41,425\,m$

$y_f = ?$

$a = 9.81\,m/s^2$

UAM

$\Delta v^2 = 2a\Delta y$

$v_f^2 - v_i^2 = 2a(y_f - y_i)$

$v_f^2 = 2ay_f - 2ay_i$

$v_f^2 + 2ay_i = 2ay_f$

$y_f = \frac{v_f^2 + 2ay_i}{2a}$

$= \frac{(367)^2 + 2(-9.81)(41,425)}{2(-9.81)}$

$y_f \approx 39600\,m$

B

$v_f = g \cdot t$

$t = \frac{v_f}{g}$

$t = \frac{367\,m/s}{9.81\,m/s^2}$

$t \approx 37.4\,s$

Δy in UAM $= 6860\,m$ RA

smaller start than Bunnganner

C

3. C jump first $t = 0$? Δy at $t = 3s$

A jump 2nd $t = 1$? $(y_c - y_a)$ at $t = 4s$

$a = -9.81$

A

$\Delta y = \cancel{v_i t}^{0} + \frac{1}{2}at^2$

$y_f = \frac{1}{2}(-9.81)(3)^2$

$y_f = -44.1\,m$

Displacement

$t = 4$

B

$\Delta y_c = 0 + \frac{1}{2}at^2$

$y_{fc} = \frac{1}{2}(-9.81)(4)^2$

$y_{fc} = -78.5\,m$

$t = 3$

$\Delta y_a = 0 + \frac{1}{2}at^2$

$y_{fa} = \frac{1}{2}(-9.81)(3)^2$

$y_{fa} = -44.1\,m$

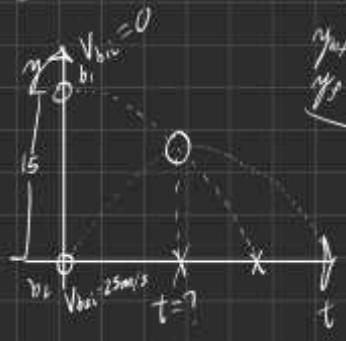
Δy

$-78.5\,m - (-44.1\,m)$

$-34.4\,m$

distance

5.



b1

$\Delta y = v_i t + \frac{1}{2}at^2$

$y_{bf} - y_{bi} = \cancel{v_{bi} t}^{0} + \frac{1}{2}at^2$

$y_f = \frac{1}{2}(-9.81)t^2 + 15$

b2

$\Delta y = v_i t + \frac{1}{2}at^2$

$y_f = 25 \cdot t + \frac{1}{2}(-9.81)(t)^2$

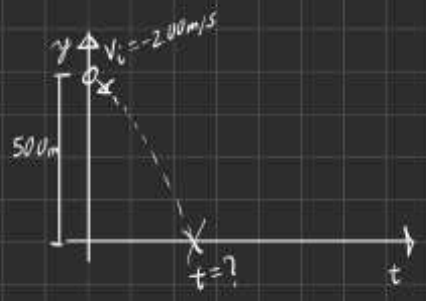
$\cancel{\frac{1}{2}(-9.81)t^2} + 15 = 25t + \cancel{\frac{1}{2}(-9.81)t^2}$

$15 = 25t$

$t = 15 \div 25$

$t \approx 0.6\,s$

6.



$\Delta y = V_i t + \frac{1}{2} a t^2$

$-500 = -2.0t + \frac{1}{2}(-9.81)t^2$

$0 = -4.905t^2 - 2.0t + 500$

$t = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

$= \frac{-(-2) \pm \sqrt{(-2)^2 - 4(-4.905)(500)}}{2(-4.905)}$

$= \frac{2 \pm \sqrt{4 + 981}}{-9.8}$

$= \frac{2 + 31.38}{-9.8}$ $= \frac{2 - 31.38}{-9.8}$

$t = -3.41$ $t = 3.00\,s$

A

$\Delta y = V_i t + \frac{1}{2} a t^2$

$-50 = V_i(2) + \frac{1}{2}(-9.81)(2)^2$

$V_i(2) = -\frac{1}{2}(-9.81)(2)^2 - 50$

$V_i = \frac{-\frac{1}{2}(-9.81)(2)^2 - 50}{2}$

$V_i = -15.2\,m/s$

B

$\Delta V = at$

$V_f - (-15.2) = (-9.81)(2)$

$V_f + 15.2 = -19.62$

$V_f = -34.8\,m/s$

C.ii

$\Delta V = at$

$V_f - (-2\,m/s) = (-9.81)(3)$

$V_f = (-9.81)(3) - (2)$

$V_f = -31.4\,m/s$

C.i

7.

$\Delta V = at$

$-V_i = (-9.81)(0.84)$

$V_i = 8.24\,m/s$

9.50 m/s

x

y

pythagorean

12.6 m/s

D

8.

Position i

$x = r\cos\theta = 5.18$

$y = r\sin\theta = 19.3$

$(5.18, 19.3)$

V_{boat} i

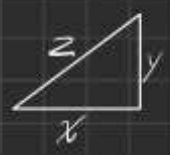
$x = r\cos\theta = 16.7$

$y = r\sin\theta = 19.9$

$(16.7, 19.9)$

$x_{ship} = 5.18 + 16.7t$

$y_{ship} = 19.3 + 19.9t$



sub-marine $z_{sub} = 50t$

Algebra :)

$(50t)^2 = (5.18 + 16.7t)^2 + (19.3 + 19.9t)^2$

$2500t^2 = 26.8 + 173t + 279t^2 + 372 + 768t + 396t^2$

$2500t^2 = 399 + 941t + 675t^2$

$0 = -1825t^2 + 941t + 399$

$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

$= \frac{-941 \pm \sqrt{(941)^2 - 4(-1825)(399)}}{2(-1825)}$

$= \frac{-941 \pm \sqrt{885481 + 2912700}}{-3650}$

$= \frac{-941 \pm 1948.9}{-3650}$

$\frac{1007}{-3650}$ ✗

$t = \frac{-2889}{-3650}$

$t \approx 0.79\,hr. \frac{60\,min}{1\,hr} = 47.5\,min$

B

Finding the ∠.

$x = 5.18 + 16.7 \cdot 0.79 = 18.4\,km$

$y = 19.3 + 19.9 \cdot 0.79 = 35.0\,km$

$\tan\theta = \frac{o}{a}$

$\theta = \tan^{-1}\left(\frac{35.0}{18.4}\right)$

$\theta = 62.3\,deg$

EofN 277 deg

A

9.

A

$V = 100 + 800 = 900$

$t = \frac{d}{v}$

$t = \frac{7700}{900}$

$t \approx 8.56\,hr$

B

$V = -100 + 800 = 700$

$t = \frac{d}{v}$

$t = \frac{7700}{700}$

$t \approx 11.0\,hr$

C

$V = -(100 \cdot \cos(30)) + 800$

$V = 713$

$t = \frac{d}{v}$

$t = \frac{7700}{713}$

$t \approx 10.8\,hr$

D

$y_i = 100 \cdot \sin(30) = 50\,km/h$

$\sin\theta = \frac{o}{h}$

$\theta = \sin^{-1}\left(\frac{50}{800}\right)$

$\theta \approx 3.58$

E

$d = r \cdot t$

$d = 50 \cdot 10.8\,hr$

$d = 540\,km$

---

10. $l = 8\,m$

$V_i = 0$

$V_f = 99.0\,mi/hr \cdot 0.447\,m/s = 44.3\,m/s$

$\Delta V^2 = 2a\Delta x$

$a = \frac{\Delta V^2}{2\Delta x}$

$a = \frac{(44.3)^2}{2 \cdot 8}$

$a = \frac{1962.49}{16}$

$a \approx 123\,m/s^2 \cdot \frac{1G}{9.81\,m/s^2} = 12.5G$

A

$V_f = at$

$t = \frac{V_f}{a}$

$t = \frac{44.3\,m/s}{123\,m/s^2}$

$t \approx 0.36\,s$

C

He might pass out

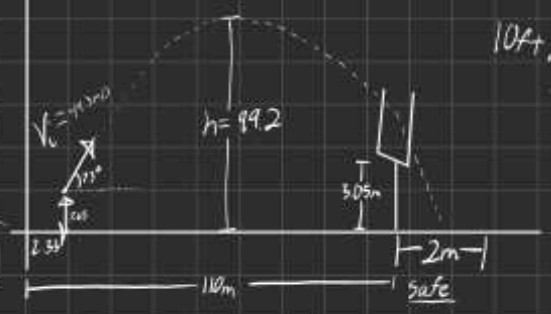
D

$t = 4.52\,s$

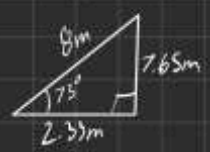
$V = 0$

$120\,yd \cdot \frac{0.9144\,m}{1\,yd} = 110\,m$

$10\,ft \cdot \frac{0.3048\,m}{1\,ft} = 3.05\,m$



cannon



$y = r\sin\theta = 7.65\,m$

$x = r\cos\theta = 2.33\,m$

$\Delta y = \cancel{V_0 t}^{0} + \frac{1}{2}at^2$

$\Delta y = \frac{1}{2}at^2$

$t = \sqrt{\frac{\Delta y}{\frac{1}{2}a}}$

$t = \sqrt{\frac{-99.2\,m}{\frac{1}{2}(-9.81)}}$

$t = \sqrt{20.2\,s}$

$t_{fall} \approx 4.50\,s$

$t_{tot} = 4.50 + 4.32$

$t_{tot} = 8.82\,s$

$x = r\cos\theta = 13.0\,m/s$

$y = r\sin\theta = 42.4\,m/s$

$d = v \cdot t$

$d = 13.0\,m/s \cdot 8.82\,s$

$d = 115\,m$

He will go past the goal post.

$\Delta V = at$

$-V_i = at$

$t = \frac{-424}{-9.81}$

$t_{up} \approx 4.32\,s$

$\Delta y = \frac{1}{2}(V_i + \cancel{V_f}^{0})t$

$y_f - y_i = \frac{1}{2}V_i t$

$y_f = \frac{1}{2}V_i t + y_i$

$y_f = \frac{1}{2} \cdot 424 \cdot 4.32 + 765$

$y_f = 99.2\,m \cdot \frac{3.281\,ft}{1\,m} = 326\,ft$

He will hit the roof.

$\Delta y = \cancel{V_0 t}^{0} + \frac{1}{2}at^2$

$y_f - y_i = \frac{1}{2}at^2$

$y_f = \frac{1}{2}at^2 + y_i$

$y_f = \frac{1}{2} \cdot (-9.81)(4.14)^2 + 99.2$

$y_f = 15.1\,m \cdot \frac{1\,ft}{0.3048\,m} = 50\,ft$

He will be outside the target area.

$\Delta x = V_i t + \cancel{\frac{1}{2}at^2}^{0}$

$110 = 13t$

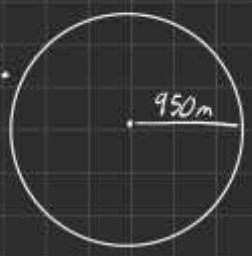
$t_{goal} = 8.46\,s$

t from top to goal $= t_{goal} - t_{up}$

$= 8.46 - 4.32$

$= 4.14\,s$

11.



$a_c = \frac{v^2}{r}$

$v^2 = a_c \cdot r$

$v = \sqrt{9.81 \cdot 950}$

$v \approx 96.5\ m/s$

$a_c = r\omega^2$

$\omega = \sqrt{\frac{a_c}{r}}$

$\omega = \sqrt{\frac{9.81}{950}}$

$\omega = 0.101\ rad/s$

$T = \frac{2\pi}{\omega} = \frac{2\pi}{0.101} = 62.2\ s \cdot \frac{1\ min}{60\ s} = 1.04\ min$

Checking $\frac{1}{2} r$

$\frac{1}{2} 9.81 = 4.91\ m/s^2$

16. $\vec{\tau} = \vec{r} \times \vec{F}$

$\tau = r \cdot F \sin\theta$

$F = \frac{\tau}{r} = \frac{5869.5\ Nm}{0.2700\ m}$

$F \approx 21740\ N$

$r = 0.2700\ m$

$\tau = 5869.5\ Nm$

$m = 2041$

$60.00\ mi/hr \cdot 0.44704\ m/s = 26.82\ m/s$

$C = 2\pi r$

$= 2\pi \cdot 0.27\ m$

$C \approx 1.70\ m$

$F = m \cdot a$

$a = \frac{F}{m}$

$a = \frac{21740}{2041}$

$a \approx 10.65\ m/s^2$

$\Delta v^2 = 2a\Delta x$

$v_f^2 = 2a x_f$

$x_f = \frac{v_f^2}{2a}$

$x_f = \frac{26.82^2}{2 \cdot 10.65}$

$x_f = 33.8\ m$

$\Delta x = \cancel{v_0 t} + \frac{1}{2} a t^2$

$t = \sqrt{\frac{\Delta x}{\frac{1}{2} a}}$

$t = \sqrt{\frac{33.8\ m}{\frac{1}{2} \cdot 10.65}}$

$t = 2.52\ s$

$33.8\ m \cdot \frac{1\ rev}{1.70\ m} = 19.9\ rev$

$a_t = r\alpha$

$\alpha = \frac{10.65}{0.27}$

$\alpha \approx 39.4\ rad/s^2$

$\tau = I\alpha$

$I = \frac{\tau}{\alpha}$

$I \approx 149\ kg\,m^2$

Activity: Conservative forces

$v_i = 2.35\ m/s$

$y_i = 0$

$g = 9.81\ m/s^2$

$KE = \frac{1}{2} m v^2$

$= \frac{1}{2} \cdot m \cdot (2.35)^2$

$KE = 2.76\ m\ J$

height

$mgy_f = 2.76\ m$

$y_f = \frac{2.76}{9.81}$

$y_f = 0.281\ m$

$PE = 2.76\ mJ$

Activity: Impulse

$I = F\Delta t = m\Delta v$

$J = \Delta p = m(v_f - v_i)$

$J = 70.0\ kg\,(5.20\ m/s - 0)$

$J = 364\ Ns$

$F_{avg} = \frac{364\ N \cdot s}{0.832\ s} = 438\ N$

Module Quiz: III.A.1

Nothing

Level 1 HW: III.A.1

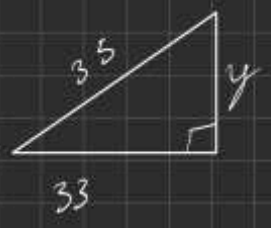
1. $m = 0.60\ kg$, $v = 20\ m/s$ — point A
$KE = 7.5\ J$ — point B

$KE = \frac{1}{2}mv^2$
$KE = \frac{1}{2} \cdot 0.60\ kg (2)^2$
$KE = 1.2\ J$

$KE = \frac{1}{2}mv^2$
$7.5 = \frac{1}{2}(0.60)v^2$
$v = \sqrt{\frac{7.5}{\frac{1}{2}(0.60)}}$
$v \approx 5.0\ m/s$

$W = KE_f - KE_i$
$7.5 - 1.2$
$W = 6.3\ J$

2.



$35^2 = 33^2 + y^2$
$y = \sqrt{35^2 - 33^2}$
$y = \sqrt{136}$
$y = 11.7\ m/s$

$\Delta v = at$
$t = \frac{\Delta v}{a} = \frac{-v_i}{a} = \frac{-11.7}{-9.81} = 1.19\ s$

$\Delta y = v_i t + \frac{1}{2}at^2$
$y_f = 11.7 \cdot 1.19 + \frac{1}{2}(-9.81)(1.19)^2$
$y_f = 13.92 - 6.95$
$y_f = 6.97\ m$

6.93

3.

$\cos\theta = \frac{a}{h}$
$a = \cos(37) \cdot 30$
$a = 24.0\ m$

$PE_i = KE_f$
$mgh = \frac{1}{2}mv^2$
$v = \sqrt{\frac{gh}{\frac{1}{2}}}$
$v \approx 10.8\ m/s$

$PE_i + KE_i = KE_f$
$mgh + \frac{1}{2}m(v_i)^2 = \frac{1}{2}m(v_f)^2$
$v_f = \sqrt{\frac{gh + \frac{1}{2}(v_i)^2}{\frac{1}{2}}}$
$v_f = \sqrt{\frac{9.81 \cdot 6 + \frac{1}{2} \cdot 16}{\frac{1}{2}}}$
$v_f \approx 11.6\ m/s$

4. $KE = 275\ J$
$p = 25\ kg\ m/s$

$KE = \frac{1}{2}mv^2$
$m = \frac{KE}{\frac{1}{2} \cdot v^2}$
$m = \frac{275\ J}{\frac{1}{2} \cdot v^2}$

$p = mv$
$m = \frac{p}{v}$
$m = \frac{25\ kg\ m/s}{v}$

$\frac{275\ J}{\frac{1}{2} v^2} = \frac{25\ kg\ m/s}{v}$
$\frac{275\ J \cdot v}{v^2} = 25 \cdot \frac{1}{2}$
$\frac{275\ J}{12.5\ kg\ m/s} = v$
$v \approx 22.0\ m/s$

$p = mv$
$m = \frac{p}{v}$
$m = \frac{25\ kg\ m/s}{22.0\ m/s}$
$m \approx 1.14\ m$

5.
$m_1 = 250\ g$ — $0.0250\ kg$
$m_2 = 100\ g$ — $0.0100\ kg$
$v_{1i} = 200\ cm/s$ — $0.200\ m/s$
$v_{2i} = 150\ cm/s$ — $0.15\ m/s$

$P_i = P_f$
$m_1 v_{1i} + m_2 v_{2i} = m_1 v_{1f} + m_2 v_{2f}$
$0.0250 \cdot 0.2 + 0.0100 \cdot 0.15 = 0.0250 \cdot v_{1f} + 0.0100 \cdot v_{2f}$
$0.005 + 0.0015 =$
$0.0065 = 0.0250(v_{2f} - 0.05) + 0.01 \cdot v_{2f}$
$0.0065 = 0.0250 v_{2f} - 0.00125 + 0.01 v_{2f}$
$0.00775 = 0.035 v_{2f}$
$v_{2f} = 0.221\ m/s$

$P_f \Rightarrow 0.200 + v_{1f} = 0.150 + v_{2f}$
$v_{1f} = v_{2f} - 0.05$

$v_{1f} = 0.221 - 0.05$
$v_{1f} = 0.172\ m/s$

6. $m_1 = 75\ kg$ $v_1 = 10\ m/s$
$m_2 = 0$ $v_2 = 0$

$m_1 v_{1i} + m_2 v_{2i} = m_1 v_{1f} + m_2 v_{2f}$
$75 \cdot 10 + 0 = 75 \cdot 5 + m_2 \cdot 5$
$750 - 375 = 5 m_2$
$5 m_2 = 375$
$m_2 = 75$

Impulse

$F_{avg} \times \Delta t = \Delta p$
$= m_2 v_{2f} - m_2 v_{2i}$
$= 75 \cdot 5 - 75 \cdot 0$
$F \times 0.1 = 375$
$F = 3750\ N$

No, bone breaking

7. $v_i = 0$ $t = 0$

$v_f = 5.20\,m/s$ $t_f \approx 0.832\,s$

Impulse

$J = |\Delta p| = m(v_f - v_i)$

$= m \cdot v_f$

$= 364\,N\,s$

$J = F_{avg} \cdot \Delta t$

$F_{avg} = \frac{J}{\Delta t}$

$= \frac{364\,N \cdot s}{0.832\,s}$

$F_{avg} \approx 438\,N$

Base

8. $m_1 v_{1i} + m_2 v_{2i} = m_1 v_{1f} + m_2 v_{2f}$

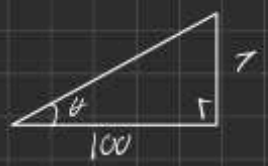
I just thought it through

Level 2 HW: Last One! ☺

1. $\frac{7\%}{100} = \frac{rise}{run}$

$0.07 = \frac{rise}{run}$

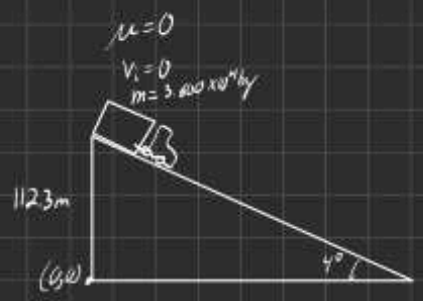
$\frac{7}{100} = \frac{rise}{run}$



$\tan\theta = \frac{o}{a}$

$\theta = \tan^{-1}\left(\frac{7}{100}\right)$

$\theta \approx 4°$



$PE_i + KE_i = PE_f + KE_f$

$mgh = \frac{1}{2}mv^2$

$v = \sqrt{\frac{gh}{\frac{1}{2}}}$

$v = \sqrt{\frac{9.81 \cdot 112.3}{\frac{1}{2}}}$

$v \approx 46.94 m/s \cdot \frac{1 mph}{0.4470 m/s} = 105 mph$

$\Delta KE = KE_f - KE_i$

$\Delta KE = \frac{1}{2}mv^2$

$= \frac{1}{2}(3.600\times10^4)(46.94)^2$

$\Delta KE = 3.970\times10^7 J$

$\Delta PE = PE_f - PE_i$

$\Delta PE = -mgh$

$= -(3.600\times10^4)(9.81)(112.3)$

$\Delta PE = -3.966\times10^7 J$

$ME = \Delta KE - \Delta PE$

0

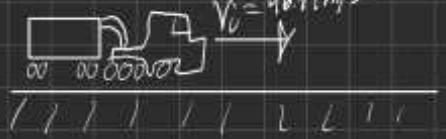
$W_{gravity} = -\Delta PE_g$

$= -(-3.966\times10^7)$

$W_g = 3.966\times10^7$

Part 3

$\mu = 0.8$



$f_f = N \cdot \mu$

$f_f = (3.600\times10^4) \cdot 0.80$

$f_f = 2.88\times10^4 N$

$F = ma$

$a = \frac{-F}{m}$

$a = \frac{-2.88\times10^4}{(3.600\times10^4)}$

$a = -0.8 m/s^2$

$\Delta V = 2a\Delta x$

$-V_i^2 = 2a(x_f - x_i)$

$x_f = \frac{-(V_i)^2}{2 \cdot a}$

$x_f = \frac{-(46.94 m/s)^2}{2(-0.8 m/s^2)}$

$x_f = \frac{-2.203\times10^3 m^2/s^2}{-1.6 m/s^2}$

$x_f \approx 1.377\times10^3 m$

Part 2

$F_N = m(\cos\theta \cdot g)$

$= (3.600\times10^4) \cdot (\cos(7) \cdot 9.81)$

$= 3.505\times10^5 N$

$f_f = \mu \cdot F_N$

$= 0.8 \cdot (-3.505\times10^5)$

$f_f = -2.804\times10^5 N$

$F_f = ma$

$a = \frac{-F_f}{m}$

$a = \frac{-2.804\times10^5}{(3.600\times10^4)}$

$a \approx -7.790 m/s^2$

$\Delta V = 2a\Delta x$

$-V_i^2 = 2a(x_f - x_i)$

$x_f = \frac{-(V_i)^2}{2 \cdot a}$

$x_f = \frac{-(46.94 m/s)^2}{2(-7.790 m/s^2)}$

$x_f = \frac{-2.203\times10^3 m^2/s}{-15.58 m/s^2}$

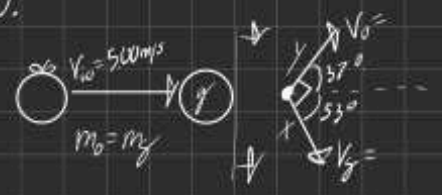
$x_f \approx 1.414\times10^2 m$

$141.4 m$

8. Made assumptions

= 3 opposite reaction.

10.



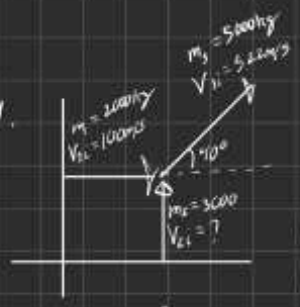
$\cos\theta = \frac{a}{h}$ $\quad$ $\cos\theta = \frac{a}{h}$

$\cos(53)\cdot 5 = a$ $\quad$ $\cos(37)\,5 = a$

$3.01 = a$ $\quad$ $3.99 = a$

X $\quad$ ✓

9.



$\tan\theta = \frac{o}{a}$

$\tan\theta \cdot a = o$

$o = 8.39 m/s$

$\vec{P}_f = m_3 + V_3 = 5000 \cdot 5.22$
$= 2.61 \times 10^4 J$

$\vec{P}_{fy} = P_f \sin(\theta_f)$
$= (2.61 \times 10^4)(\sin(40))$
$= 1.68 \times 10^4$

$P_{fy} = m_2 + V_{2i}$

$V_{2i} = \frac{P_{fy}}{m_2}$

$V_{2i} = \frac{1.68 \times 10^4}{3{,}000}$

$V_{2i} \approx 5.59 m/s$

---

14. $35 mph \cdot \frac{0.44704 m/s}{1 mph} = 15.65 m/s$

$\Delta V = at$

$15.65 m/s = a \cdot 0.09 s$

$a \approx 174 m/s^2 \cdot \frac{1G}{9.81 m/s^2} = 17.7G$

18. $a_{avg} = \frac{\Delta V}{\Delta t}$

$\Delta V = a_{avg} \cdot \Delta t$

$-V_i = 174 \cdot 0.09$

$V_0 = -15.7 m/s$

$\Delta \vec{p} = m\Delta \vec{V}$
$= 5 \cdot (-15.7)$
$= 78.3 N$

$\vec{J} = \vec{F}_{avg} \cdot \Delta t$
$= 870 N \cdot 0.09 s$
$= 78.3 N \cdot s$

yes

$F = m \cdot a$
$= 5 \cdot 174$
$F \approx 870 N$

---

19. $\Delta V^2 = 2a\Delta x$

$-V_i^2 = 2a x_f$

$a = \frac{-V_i^2}{2x_f}$

$a = \frac{-(26.82)^2}{2(27.43)}$

$a \approx -13.1 m/s^2$

$60 mph \cdot \frac{0.44704}{1 mph} = 26.82 m/s$

$90 ft \cdot \frac{0.3048 m}{1 ft} = 27.4 m$

$\Delta V^2 = 2a\Delta x$

$V_f^2 - V_i^2 = 2a x_f$

$-V_i^2 = 2(-13.1 m/s^2)(160 m) - (8.94 m/s)^2$

$V_i = \sqrt{-499 m^2/s^2}$

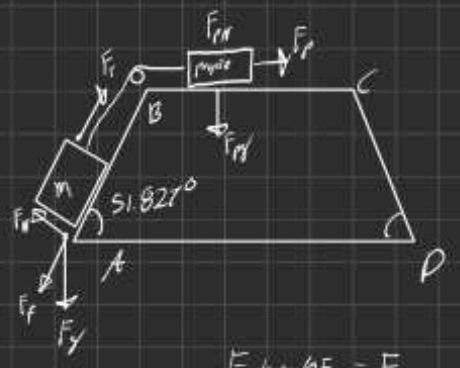
$V_i = 22.3 m/s \cdot \frac{1 mph}{0.44704 m/s} \approx 50.0 mph$

$20 mph \rightarrow 8.94 m/s$

$52.5 ft \rightarrow 16.0 m$

$\Delta x =$

26.



Lime on Lime $\mu = 0.7$

Lime on wet sand $\mu = 0.350$

$m = 2.50\,tons = 2268\,kg$

$$F_f + \cos\theta F_g = F_T$$

$$N \cdot \mu + \cos\theta F_g = F_T$$

$$\sin\theta F_g \cdot \mu + \cos\theta F_g = F_T$$

L.L

$$\sin(141°) \cdot m \cdot a \cdot \mu + \cos(141°)\, m \cdot g$$

$$0.618 \cdot 2268 \cdot 9.81 \cdot 0.7 + (-0.777)\, 2268 \cdot 9.81$$

$$9624 + 17291$$

$$26916\,N = F_T$$

$$F_T = 2.69 \times 10^4\,N$$

$$\downarrow$$

$$6050\,lbf$$

$$6.05 \times 10^3\,lbf$$

L.W

$$\sin(141°) \cdot m \cdot a \cdot \mu + \cos(141°)\, m \cdot g$$

$$0.618 \cdot 2268 \cdot 9.81 \cdot 0.350 + (-0.777)\, 2268 \cdot 9.81$$

$$4812 + 17291$$

$$22100 = F_T$$

$$F_T = 2.21 \times 10^4\,N$$

$$\downarrow$$

$$4970\,lb\text{-}f$$

$$4.97 \times 10^3\,lb\text{-}f$$

force per person = weight · 0.5 $\quad$ 165 lbf → 734 N

= 734 N · 0.5

= 367 N per person

L.L

$$26916\,N \cdot \frac{1\,person}{367\,N} = 73.3\,people$$

L. wet

$$22100\,N \cdot \frac{1\,person}{367\,N} = 60.2\,people$$